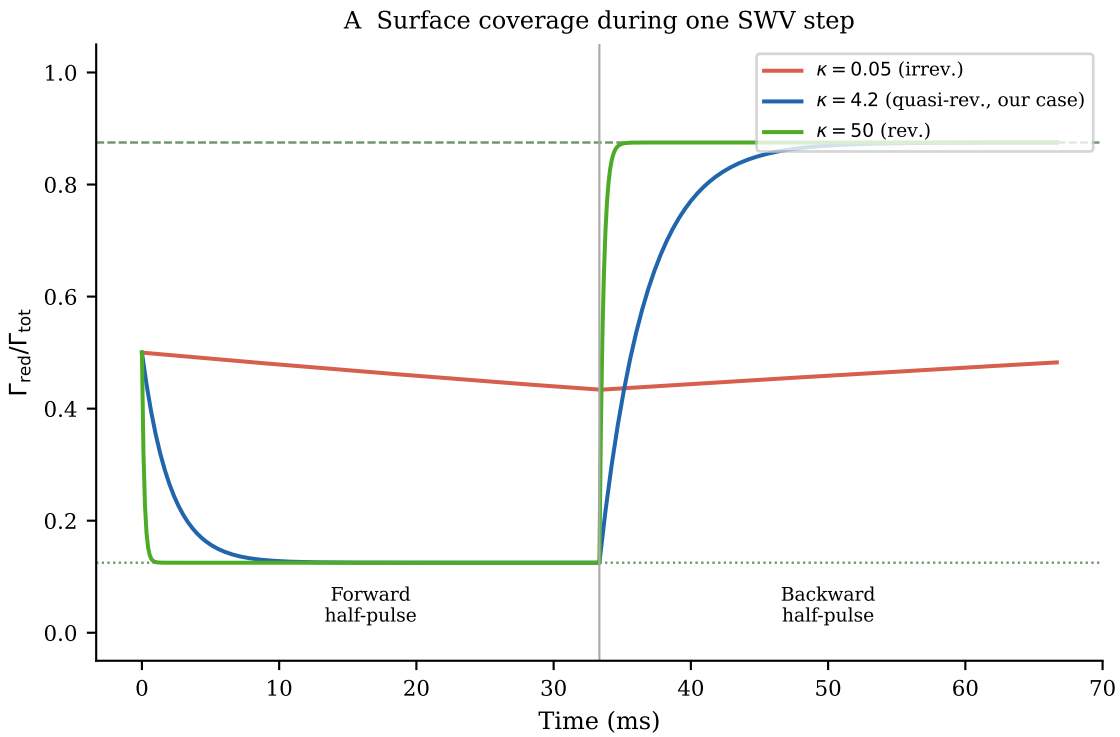


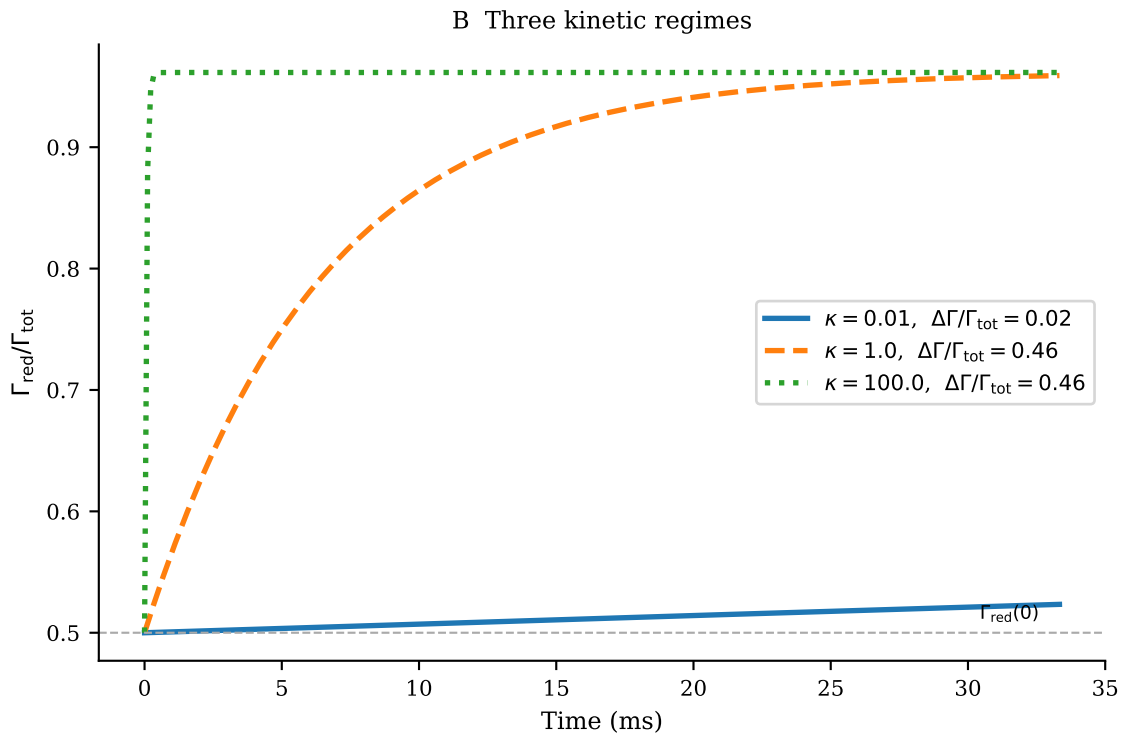


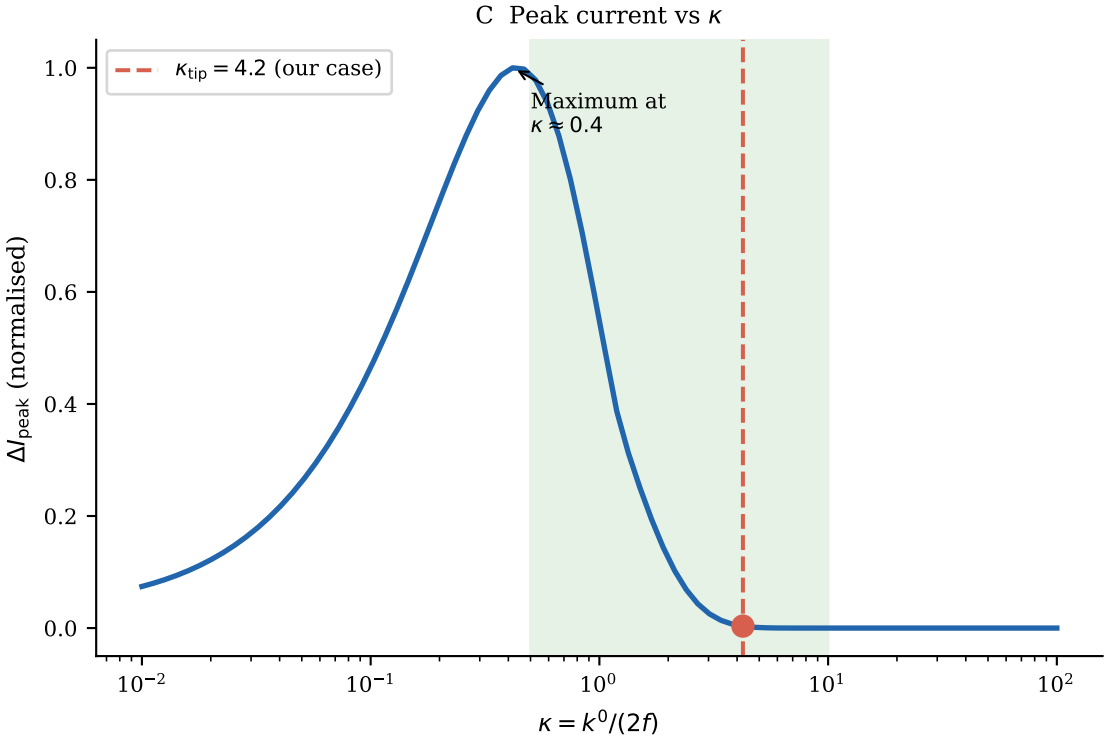


D Integrating factor derivation

Integrating factor derivation:

ODE: $\frac{d\Gamma_{red}}{dt} = k_f\Gamma_{tot} - (k_f + k_b)\Gamma_{red}$

Let $a = k_f\Gamma_{tot}$, $b = k_f + k_b$:

$\dot{y} + b\,y = a$

Multiply by $\mu = e^{bt}$:

$\frac{d}{dt}[y\,e^{bt}] = a\,e^{bt}$

Integrate, apply IC $y(0) = \Gamma_{red}(0)$:

$$\boldsymbol{\Gamma_{red}(\tau) = \Gamma_{eq} + [\Gamma_{red}(0) - \Gamma_{eq}]\,e^{-b\tau}}$$

where $\Gamma_{eq} = k_f/(k_f + k_b)\,\Gamma_{tot}$